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Carbon tax, energy policy, and sustainable development in Indonesia

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Abstract

The effectiveness of carbon taxes has been subject to important debates on the sustainable development pathway. As an archipelagic country that is vulnerable to climate change, the impact of implementing this policy instrument is still understudied in the literature. This study examines the effect of carbon tax and recycling scenarios on the Indonesian economy by employing a dynamic computable general equilibrium (CGE). The result revealed that carbon tax revenue recycling combined with household transfer and investment yielded better GDP and employment rate than a household transfer only in the medium and long terms. Furthermore, the carbon tax revenue recycling on household transfer and renewable energy investment created energy intensity reductions. This study recommends implementing the tax by prioritizing specific sectors, setting appropriate tax levels, applying a transitional policy, and diversifying the investment strategy to align with the sustainable development pathway.

KEYWORDS

carbon tax, CGE model, emission mitigation, energy intensity, revenue recycling, sustainable development

1 | INTRODUCTION

The 2030 United Nations Agenda for Sustainable Development and the Paris Agreement have a strong relationship in providing a global blueprint for sustainable development related to clean energy and sustainable life. It manifests both issues through the 7th and 13th sustainable development goals (SDGs), namely affordable and clean energy and climate action. Thus, the Paris Agreement aims to combat climate change, keeping a global temperature rise below 2°C (3.6 °F) and pursuing efforts to limit temperature rise to 1.5°C (2.7 °F). It also aims to strengthen the countries' ability to deal with the impacts of climate change (United Nations, 2018). A sustainability pathway that poses the least climate change challenges is more conducive to increasing well-being from an inclusive wealth framework (Kurniawan & Managi, 2018). The Paris Agreement requires each member country to submit its national climate plan, called Nationally

Determined Contributions (NDCs), to meet the targets. Several countries set emission and energy consumption reduction and increased renewable energy utilization targets to achieve the SDG's goals and comply with the Paris Agreement. Each country also sets strategies to boost renewable energy utilization by increasing renewable energy investment, improving energy efficiency, and reducing greenhouse gas emissions, directly affecting climate change. Hence, achieving SDG targets on energy will enhance compliance with the Paris Agreement.

Several years after the Paris Agreement, global carbon emissions increased by 2.4% and 0.9% in 2018 and 2019, respectively (Olivier & Peters, 2020). Under current policies, the world is predicted to warm by 3.0°C at the end of the century (Climate Action Tracker, 2019). The number is twice that of the Paris agreement target. Also, investment in renewable energy declined by 11% in 2018 and slightly increased by 2% in 2019. Those investments were \$325 billion,

\$288.3 billion, and \$301.7 billion in 2017, 2018, and 2019, respectively (Ajadi et al., 2020). Though the global progress of the Paris Agreement is lagging, there might be some variation in the individual countries' achievement that needs to be further assessed.

Considered one of the growing emitters, Indonesia is also one of the most vulnerable countries to climate change. As an archipelagic country, a significant proportion of 250 million people spread across thousands of islands and low-lying coastal areas. Having such a condition, Indonesia provides a good case study. The country has committed to achieving SDG's target by increasing the renewable energy mix supply and doubling its primary energy efficiency. The Government of Indonesia (Gol) has established the General Plan of National Energy—*Rencana Umum Energi Nasional* (RUEN)—by reducing energy intensity by up to 1% annually and achieving 23% of renewable energy (RE) mix by 2025 (RUEN, 2017). Responding to the Paris Agreement, the Gol, coordinated by the Ministry of Environment and Forestry, issued law 16/2016 of Indonesia national determined contribution (INDC) targets of national emission reduction per sector. According to the NDC targets, Indonesia commits to reducing emissions by up to 29% in 2030 and up to 41% under the conditional scheme. It needs to reduce CO₂ emissions by about 18.8% (Direktorat Jenderal Pengendalian Perubahan Iklim, 2019). The policy is also in line with the government's efforts in preparing an NDC update and a Long-Term Strategy for Low Carbon and Climate Resilience in 2050 (Ministry of Environment and Forestry, 2021). Furthermore, the Ministry of Planning, through the Medium-term National Development Plan (RPJMN) 2020–2024, targets greenhouse gas emissions of around 27.3%, lower than business-as-usual (Climate Action Tracker, 2020). The Gol needs to act ambitiously by setting up more inclusive and tighter policy strategies to meet the RUEN's and INDC's targets.

Among the options available in the literature to promote clean energy and energy mix improvement, fiscal policy, including removal of subsidy or fuel tax, carbon tax, government spending, and or investment, is widely used around the globe. The carbon tax is expected can reduce carbon emissions by enhancing innovation and energy efficiency (Ulucak & Kassouri, 2020). Moreover, a carbon tax is considered superior to other fiscal instruments, given its impact on emission reduction and its easiness of implementation (Zhou et al., 2018). The rational aspect is that people will change their consumption behavior toward more efficient and greener energy by imposing a carbon tax. Then, the tax revenue could be invested in green energy or infrastructure development. Carbon taxes are also considered essential drivers for improving ecological quality (Ullah et al., 2022). Thus, a carbon tax could reduce emissions, potentially increase the RE mix, or even improve the economy and environment.

Having such importance, only a few studies examine the impact of the carbon tax policy in Indonesia. A carbon tax policy is predicted could reduce CO₂ emissions in the country (Ayu, 2018). Focusing on the distributional impact of a carbon tax, it is estimated that the intervention in Indonesia is not necessarily regressive (Yusuf & Resosudarmo, 2015). There are different perspectives concerning the implementation of the policy. Policymakers tend to be open to policy intervention, while the business community opposes a carbon tax due

to its influence on productivity and competitiveness (Dyarto & Setyawan, 2021). The previous study focuses on the impact of the carbon tax on the economy and pollution reduction (Baranzini et al., 2000; Yusuf & Resosudarmo, 2015), but a few studies debating the impact of the carbon tax on energy consumption, particularly in developing countries. Theoretically, the higher price of energy due to carbon tax could induce energy demand saving on energy, especially for brown energy, and create energy efficiency. Suppose RE is available at a more affordable price. In that case, the demand for RE could increase, and the total energy consumption might be stable or slightly increase compared to the non-existing RE. The economic impact of energy consumption is not necessarily different, but the impact on emission reduction could be substantial. Hence, our study complements the existing literature discussing the impact of the carbon tax on energy consumption. Furthermore, our study also examines the impact of the carbon tax on economic aspects, such as GDP, real consumption, employment rate, and sectoral output, and environmental indicators, such as emission and energy intensity, to make it more comprehensive. The previous study still did not answer the impact of implementing this policy instrument on the NDC target of Indonesia and the policy roadmap in the energy sector.

Against the backdrop, this paper attempts to investigate the impact of a carbon tax on achieving the SDGs and Paris Agreement targets manifested in the RUEN's and INDC's targets. Besides, it also estimates the effect of a carbon tax on environmental and macroeconomic indicators using a recursive dynamic CGE model. The CGE model has been widely used as the primary method in assessing the economy-wide impact of the energy-fiscal instrument, including carbon tax (Hartono, Hastuti, et al., 2020; Hartono, Komarulzaman, et al., 2020; Holmøy, 2016; Yusuf & Resosudarmo, 2015). Four carbon tax implementation (with different recycling combinations) scenarios were simulated in a quest for a better policy option. These simulations included (1) levy carbon tax only, (2) levy carbon tax and distribute all the carbon tax revenue to households, and lastly, levy carbon tax and distribute half of the revenue to households and half for infrastructure (3a) or renewable energy investment (3b).

This article may contribute to three aspects. Firstly, it adds to the existing literature on the impact of a carbon tax by employing a dynamic CGE model with detailed energy sectors, in contrast to the aggregated data on energy in the standard input–output (IO) table. Also, the detailed energy sectors differentiate this study from previous ones (Antimiani et al., 2014; Ojha et al., 2020; Tvinnereim & Mehling, 2018), allowing a better analysis of the environmental impact of a carbon tax, primarily the carbon emission, energy intensity, and energy mix. Furthermore, the dynamic CGE model could capture market reactions and structural adjustments more comprehensively, for instance, by allowing endogenous adjustment in the supply of factors and technology (Anderson, 2020). Secondly, the study provides a comprehensive simulation of emission and energy intensity by levying and recycling the carbon tax on households' lump-sum transfer, infrastructure, and renewable energy investment. In contrast to other studies focusing more on macroeconomics and household consumption changes (see Yusuf & Resosudarmo, 2015; Li et al., 2020).

The present research is essential as Indonesia is listed as the top 10 largest emitter countries. Hence, its energy mix changes may significantly affect the global environmental condition, including the achievement of SDGs and the Paris Agreement.

The rest of this article is organized as follows. Section 2 provides the concept and evidence of fiscal instruments in promoting RE and energy efficiency, and Section 3 presents Indonesian government policies and strategies to achieve RUEN's and NDC's targets. Section 4 covers the method and data sources, followed by Section 5, describing the result and discussion. The last section, Section 6, contains the conclusion and policy implications.

2 | THE FISCAL INSTRUMENT IN ENVIRONMENTAL AND ENERGY POLICY: CONCEPT AND EVIDENCE

2.1 | Carbon tax as a fiscal instrument

Environmental tax, such as carbon tax, could play a vital role in achieving SDG goals related to clean energy and sustainable life. This policy has been considered an effective tool for developing a low-carbon economy (Zhang & Zhang, 2019). Earnings for environmental and social purposes could be brought about, moving behavior toward low-carbon activities. Also, green investment through pricing environmental externalities could be triggered (Carraro et al., 2015). The tools mentioned earlier may shape energy prices by foisting a tax to reduce energy consumption or financial support (when first cost-related barriers are addressed). Consequently, taxes are considered to be the best tools for environmental economists, as they can provide a uniform signal to the whole economy and equalize compliance costs (Koeppel & Ürge-Vorsatz, 2007).

Putting a price on carbon is crucial for reducing carbon emissions, decreasing carbon intensity, and stimulating green growth (Antimiani et al., 2014; OECD, 2013), so any plan for low-carbon growth should be based on it. There are several ways to determine carbon price emissions: carbon pricing, internal carbon pricing, and implicit carbon pricing. Carbon pricing is an energy policy tool that captures GHG emissions' external costs and expresses it in a monetary unit per tCO₂e. The carbon price provides an economic signal to emitters, allowing them to reduce their emissions or continue to emit by paying for them. Carbon pricing includes an emissions trading system (ETS), carbon tax, offset mechanism, and Results-based climate financing (RBCF) (World Bank, 2019b). Meanwhile, internal carbon pricing refers to a company's value voluntarily for internalizing its green economy cost of GHG, that is, shadow price and internal carbon tax (Institute for Climate Economics, 2016). At last, implicit carbon includes removing fossil fuel subsidies, energy taxation, fuel taxation, support for renewable energy, energy efficiency certificate trading, and regulatory standards (OECD, 2013; World Bank, 2019b).

Experts and international organizations, for example, United Nations (UN), Organization for Economic Co-operation and Development (OECD), and European Union (EU), have strongly recommended

the carbon tax as the most efficient instrument to reduce carbon emissions (Baranzini et al., 2000; Cuervo & Gandhi, 1998; Ojha et al., 2020; Timilsina, 2018; Tvinnereim & Mehling, 2018). The carbon tax can raise revenue while keeping the carbon at a certain threshold. Also, the implementation of a carbon tax could contribute to achieving market efficiency, minimizing environmental impact, and raising domestic resources to fund public goods. When implementing a carbon tax, it is necessary to consider revenue use, tax rates, and adverse effects, such as consumption and employment changes. Carbon tax revenue may be allocated to finance government spending or recycled through subsidies, household transfers, investment, or other tax reductions. Both actions will affect economic activities at the micro and macro levels. When levying the tax, the government may start with gradual increases or a one-shot increase in tax rates, both of which have positive and negative aspects. Lower tax may be more politically acceptable, but lower revenue to spend or recycle. In contrast, a one-shot tax may not be favorable but can socially optimize and cut unwanted activities via price signaling. Furthermore, a carbon tax without other affirmative policies, such as household transfer or labor tax reduction, may hamper household consumption and employment rate (World Bank, 2019a).

However, the carbon tax policy risks economic growth by raising energy prices. The increase in energy prices could hamper energy consumption and welfare. The effect is apparent in countries that rely on fossil fuels, like Indonesia. Lower-income households who use fossil fuels excessively might suffer more from the carbon tax policy. On the other hand, higher energy prices might encourage households to use energy efficiently or substitute alternative energies.

Until now, Indonesia has not implemented a carbon tax policy, but previous research has simulated the carbon tax implementation in Indonesia. For instance, Dissanayake et al. (2020) evaluated carbon tax, fuel tax, and ETS for Indonesia. They find that fuel tax increased economic growth by 0.29% by 2030, but carbon tax and ETS reduced economic growth by about 0.11%. But the distraction could be minimized by applying revenue-recycling from carbon tax via a uniform reduction in the commodity tax rate (Yusuf & Resosudarmo, 2015). Nevertheless, Indonesia implemented an implicit carbon tax policy, that is, fossil-fuel energy subsidy removal, in 2015. At the same time, the policy reduces energy consumption, generates output contraction, especially in the transportation and manufacturing sectors, and generates demand-side savings in annual electricity use (Burke & Kurniawati, 2018; Yusuf et al., 2017).

There have been many studies examining various policies related to the environment. Among the quantitative analysis, CGE modeling is the primary method employed in carbon tax research. For example, Antimiani et al. (2014) used the CGE model GDynE to examine the options between economic growth and low carbon targets in developing and developed countries up to 2035 through the green climate fund. They estimate by simulating two main scenarios, i.e., a scenario for meeting the 450 PPM target (quota) with national sub-targets (meet the domestic target) without international trading and a similar scenario with global carbon market trading. The second scenario seems more efficient and could reduce the cost by around 1% of GDP compared with the first one. With international global trading, a country could have more funding and reinvest the fund in renewable

energy and more energy-saving sectors. Next, Landa Rivera et al., 2016 simulated the medium and long-term impacts of the proposed and expected energy policy on the environment and the Mexican economy using the Multi-sector Macroeconomic Model for the Evaluation of Environmental and Energy Policy (Three-ME) (Landa Rivera et al., 2016). The result showed that under total redistribution of carbon tax revenues, a double dividend arises: the policy appears beneficial in GDP and CO₂ emissions reduction. Afterwards, Timilsina (2018) summarized various carbon tax studies for 30 years. The paper presents the economic impacts, revenue recycling choices, distributional implications and competitiveness, and border tax adjustment related to the carbon tax.

In Indonesia, several studies are focusing on the CGE to investigate the possibility of emission reduction in various ways. Syaifudin et al. (2015) analyzed the impact of financial support on the sub-national region to implement an energy efficiency policy in Indonesia (Syaifudin et al., 2015). They used CGE Model (IRSA Indonesia 5) and segregated Indonesia into five regions, that is, Sumatera (R1), Java-Bali (R2), Kalimantan (R3), Sulawesi (R4), and East Indonesia (R5). By implementing several scenarios, they suggested that the transfer should be distributed lump-sum to be more effective in the emission reduction efforts. Furthermore, Hartono, Hastuti, et al. (2020), Hartono, Komarulzaman, et al. (2020) examined the effect of the current subsidy reform on energy mix targets and energy intensity using the recursive dynamic CGE model. They showed that eliminating the energy subsidies is insufficient to achieve the energy mix and intensity target. Another research conducted by Dissanayake et al. (2020) evaluated carbon tax, fuel tax, and ETS for Indonesia. The results showed that fuel tax increased economic growth by 0.29% by 2030, but the carbon tax and ETS reduced economic growth by about 0.11%. Then, the research suggested that the Gol needs to levy a carbon tax of around US\$36/ton CO₂ to meet the emissions targets. However, Yusuf & Resosudarmo (2015) revealed that compared to the lump-sum transfer, revenue-recycling from carbon tax via a uniform reduction in the commodity tax rate might reduce the contraction of aggregate output due to carbon tax implementation. Also, Malahayati and Masui (2019) explored the impact of mitigation policy in the Land Use Change and Forestry (LUCF) sector on the economy and environment. They found that increasing the crop yield and conserving the land together could reduce emissions from LUCF and protect forest areas (Malahayati & Masui, 2019). Siagian et al. (2017) employed The Asia-Pacific Integrated Model/CGE (AIM/CGE) to simulate GHG emission through low-carbon energy technologies. They revealed that RE and natural gas utilization in the power and transport sectors could achieve a low-carbon energy system. Also, the electrification of end-user consumption supplied by renewable power generation supports the system (Siagian et al., 2017).

3 | OVERVIEW OF INDONESIA'S ACTIONS ON ENERGY POLICY

Based on the Climate Change Performance Index (Burck 2021, Indonesia ranks 24 among 57 countries and the European Union (EU),

which are together responsible for more than 90% of Global Greenhouse Gas (GHG) emissions. Compared to the CCPI 2020, Indonesia remains the medium performer in climate protection (Burck, 2021). World Resources Institute (Wijaya et al., 2017) stated that Indonesia is the world's eighth-largest emitter of greenhouse gases and the most significant contributor to forest-based emissions (Wijaya et al., 2017). According to The MEF, in 2018, Indonesia produced 1637 million metric tons of carbon dioxide equivalent (MtCO₂e) emissions for the three leading greenhouse gases: carbon dioxide (CO₂), methane (CH₄), and nitrogen dioxide (NO₂) (Kementerian Lingkungan Hidup dan Kehutanan, 2020). Indonesia's greenhouse emissions have risen by 450,9 MtCO₂e compared to 2000 and are predicted to increase even faster over the next 15 years under a business-as-usual (BAU) scenario. Increased emissions from the energy and land-use sectors drove this increase. In 2018, 36,4% or 595.7 MtCo2e emissions were produced from energy-related activities, comprising fuel combustion and fugitive emissions from fuel production. Most (94.1%) of the energy-related emission was in the form of carbon gas (CO₂) (Boer et al., 2018).

According to the INDCs, Indonesia voluntarily pledged to reduce emissions by 26% and 29% in 2020 and 2030 under the BAU scenario (up to 41% conditional on international support). Indonesia is on track to meet its 2020 target but needs more effort to meet the 2030 target as the energy sector heavily relies on fossil fuels. The GHG emissions from this sector are projected to more than double by 2030 (OECD, 2019). Besides, Indonesia's 2014 National Energy Policy predicts the use of coal nearly double by 2025 (compared to 2015 levels) to achieve adequate supply for all. In contrast, Indonesia's total renewable energy in primary energy supply (excluding biomass) was only 9.18% in 2019 (Ministry of Energy and Mineral Resources Republic of Indonesia, 2020), still far from the target (23%) of energy supply in 2025.

The accumulation of GHG will have an impact on climate change. Indonesia's geographical and socioeconomic conditions make it vulnerable to natural disasters, including extreme weather and climate change (Kementerian Lingkungan Hidup dan Kehutanan/Ministry of Environment and Forestry, 2019). The USAID (2016) stated that the potential economic loss could reach IDR 132 trillion by 2050 (USAID, 2016). The OECD (2019) also projected that overall climate change damage would reach about 2.3% of GDP by 2060. Realizing this damage, Indonesia ratified Paris Agreement in 2016. In response to the agreement, the Gol, coordinated by MEF, issued law 16/2016 as a guide to achieving Indonesia's National Determined Contribution (INDC) targets of national emission reduction per sector. Previously, the Gol issued an action plan to reduce GHG in 2011 (Rencana Aksi Nasional Penurunan Emisi Gas Rumah Kaca, RAN GRK). In 2014, the Gol produced Gol Regulation 79/2014 on National Energy Policy (Kebijakan Energi Nasional, KEN). The three main objectives in KEN are to increase energy access and security, increase energy efficiency, and reduce energy elasticity so that the value is less than 1 (Patunru & Rakhmah, 2017). Then, based on the KEN's objectives, the Gol issued the General Plan of National Energy (RUEN) through Presidential Regulation 22/2017. Simultaneously, the Ministry of Energy and Mineral

Resources (MEMR) prepared the General Plan of National Electricity (RUKN). RUEN provides policy on the national energy management plan and acts as an elaboration and implementation plan of KEN until 2050, while RUKN offers a general plan for developing the electricity supply system (DEN, 2019).

Efforts to decarbonize the energy sector need to accelerate. The mitigation focus has been on fuel switching, energy efficiency, and public transport (Kementerian Lingkungan Hidup dan Kehutanan/Ministry of Environment and Forestry, 2019). The Gol has carried out several mitigations to address climate change issues from energy sectors, including taxes, charges, and prices, Feed-in Tariff (FiT), and promotion of Renewable Energy (RE) power plants. Revenue from motor vehicle fuel tax and a local street lighting tax amounted to 0.2% of Indonesia's GDP (OECD, 2019). The Gol is still working on increasing the fuel tax rate and reducing fuel subsidies. As many studies have shown that energy subsidy in Indonesia was misplaced, regressive, mistargeted, and unproductive, hindering the promotion of energy-saving and countering attempts to reduce GHG emissions (Burke & Resosudarmo, 2012; Patunru & Basri, 2012; Yusuf & Resosudarmo, 2011; Yusuf & Sumner, 2015).

The MEMR introduced the FiT agreement in July 2016, Regulation 19/2016. Under the FiT scheme, the Gol compensated independent power producers (IPPs) a fixed and above-market price to buy RE. The agreement applies from around 20 until 30 years. In 2017, the Gol replaced the FiT with a new tariff mechanism and set the maximum rate based on the regional electricity cost. The Gol also tried accelerating biofuel energy by issuing Presidential Regulation 61/2015. Furthermore, the MEMR Regulation 53/2018 is supported by adding liquid biofuels as part of RE's source for electricity generation.

Although there have been many studies in various countries on the effectiveness of carbon tax in reducing emissions, the Indonesian government has not yet issued a carbon tax law. However, the Gol has issued Presidential Regulation No. 77/2018 as a legal framework regulating carbon trading and receiving international grants or funds for emission reduction.

4 | METHODOLOGY

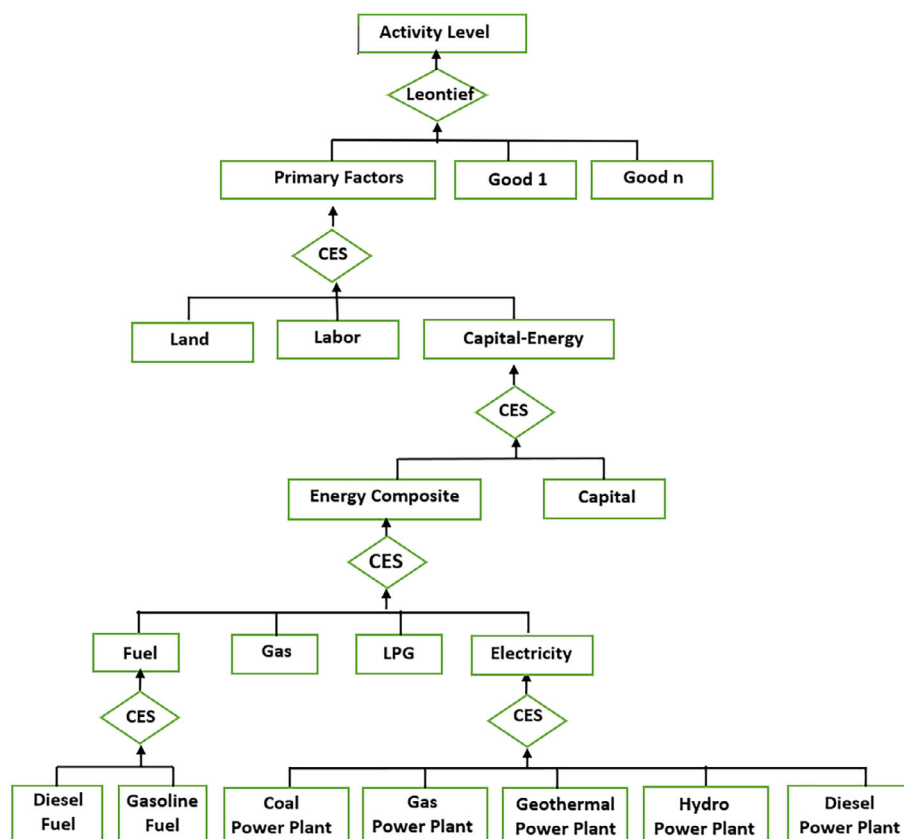
Several methods are frequently used to determine the effects of the carbon tax, including microsimulation (Saelim, 2019), input-output analysis (Renner, 2018), and CGE (Saelim, 2019; Yusuf & Resosudarmo, 2015; Zhou et al., 2018). Among the tools accessible for energy policy analyses, the CGE model was comprehensively accepted as a proper empirical tool (Holmøy, 2016). The CGE models (Taylor, 2016) can pinpoint the economic system's responses deriving from changes in energy policies such as carbon taxes, which are market-based actions. In the dynamic version, CGE can simulate the effects on the long-term growth of variations in the rate of capital accumulation that might result from changes in investment due to modified saving behaviors (Jorgenson & Yun, 2013). Moreover, the CGE model is highly disaggregated, which allows for examining

income distribution among household groups and industry-specific impacts. Therefore, the CGE model is a compelling tool to simulate the effects of fiscal or economic policy changes (Quatrebarbes et al., 2016; Lemelin and Savard, 2022).

This analysis employed a dynamic CGE model of a small open economy based on the ORANIG-RD (Horridge, 2002). Computable or applied general equilibrium models are founded on Walrasian general equilibrium theory (Walras, 1874; Wald, 1936; Arrow and Debreu, 1954; Arrow and Hahn, 1971) and, frequently—in an intertemporal context—the theory of overlapping generations (Allais, 1947; Samuelson, 1958; Diamond, 1965; Modigliani, 1966). CGE is an economic model that represents the entire national economy but aggregates microeconomic behavior in great detail. The model is represented by a system of n nonlinear equations containing n endogenous variables and a large number of exogenous variables. The system of equations determines commodity prices and input quantities (including primary inputs such as labor, capital, and land, as well as intermediate inputs). In a competitive market economy, optimizing rational economic agents, in this case producers and consumers, are represented by the CGE model's specified equations. These make up the market-clearing demand and supply for commodities, represented in the model as the market-clearing conditions or equilibrium. Compared to the static model, the primary advantages of this newer generation of CGE models are their capacity to account for lifecycle behavior, intergenerational issues, intertemporal investment and consumption decisions, and government deficits. The dynamic general equilibrium dynamic model is characterized by three significant mechanisms: (1) the stock-flow relation between investment and capital stock (assuming a one-year gestation lag), (2) the positive relation between investment and profit rate, and (3) the relationship between wage growth and employment.¹

This study adjusted the standard ORANIG-RD model² to accommodate energy policy analysis. The adjustment involved the energy substitution probabilities and the inclusion of carbon emissions. Energy materials are considered intermediate inputs with the Leontief production function in the standard model; consequently, the model does not permit price-induced substitution, leading to the adjustment in the energy substitution probability and the inclusion of carbon emission. Firstly, the study altered the model to allow substitutions among commodities (inter-energy substitution) and substitution between capital and energy (capital-energy substitution), following (Wianiwat & Asafu-Adjaye, 2013). Secondly, the carbon tax equation was added following MMRF-Green by (Adams et al., 2002). The production function is regarded as separable, following the nested structure (see Figure 1). The intermediate inputs, a composite of primary factor and energy, were combined in a fixed proportion of the Leontief production function to produce commodities at the top level. The constant elasticity of substitution (CES) production function, at the lower level, provides the aggregation of domestic and imported intermediate inputs, following the Armington assumption in which the producer minimizes the costs through the cheapest inputs from domestic or imported sources (Armington, 1969). The cost of primary input (labor, land, and capital-energy) composite demand is minimized using a CES function.

FIGURE 1 Production structure of energy sectors. Source: Horridge (2002), modified by authors



The inter-substitution between capital and energy was determined using a method by Wianwivat and Asafu-Adjaye (2013) in the following stage. In the long run, the inter-substitution between capital and energy was accommodated by utilizing the CES function. The present study established the energy composite at the lower stage by combining energy inputs (using the CES function) at two nesting stages to reduce production costs. In the initial phase, energy input was nested based on types, that is, (i) Fuel, (ii) Electricity, (iii) LPG, and (iv) Gas. In the following phase of energy nesting, fuel was a composite of Diesel and Gasoline-type fuel. Meanwhile, a composite of generator-type energy was electricity, encompassing coal, gas, geothermal, hydro, and diesel power plants. The energy nesting appealed to the households' demand for energy.

Meanwhile, the household commodities demand was seized in a linear expenditure system (LES) following Yusuf (2015). This function is a popular option for modelling consumer preferences in CGE (Boer et al., 2021). The mentioned function was purposively used in the composite of both imported and domestic commodities. Meanwhile, the combination of domestic and imported goods was designed utilizing the Armington aggregation, similar to the construction of a composite of intermediate input (Yusuf, 2015). For each commodity, the household was assumed to determine the optimal combination between domestic and imported goods while keeping down the total expenses for a given amount of the composite. Next, each household augments the Stone-Geary utility function to reach the optimum

commodities satisfying the budget constraint. This utility function distinguishes subsistence from the household demand's luxury component.

4.1 | Database

The latest available 2010 Indonesian Input-Output (IO) Table and the updated 2008 Social Accounting Matrix (SAM) from Statistics Indonesia (Badan Pusat Statistik/BPS) were consolidated to form the database. In allowing more details on the energy and electricity sectors, the 2010 standard I-O from BPS (comprising 185 sectors and commodities) was extended into 191 sectors and commodities. Diesel fuel, gasoline fuel, coal, gas, and LPG were inserted in the energy sectors. In contrast, the study disaggregated the electricity sector by the generator's type, involving coal, Gas, Geothermal, Hydro, and Diesel power plants. Next, the mentioned sectors were disentangled into 63 sectors following the objectives of the study.

Meanwhile, the present study used Indonesia SAM information to capture the income transfer from/to various economic agents to govern the inter-energy substitution. It employed parameters value estimated by Adams et al. (2002) and Benavente (2016). The model will first run a baseline simulation followed by policy simulations to perform a scenario using a dynamic model. The baseline simulation illustrates the business as usual, while the policy simulations demonstrate the economy's different path if policy changes are applied. The

TABLE 1 Scenarios mapping.

Simulation	Shock
SIM1	Carbon Tax USD 28.8/Ton of CO ₂
SIM2	SIM1 + lump sum transfer (100%)
SIM3a	SIM1 + lump sum transfer (50%) + infrastructure investments (50%)
SIM3b	SIM1 + lump sum transfer (50%) + renewable energy investments (50%)

baseline simulation needs historical macroeconomic data between (base year) 2010 and 2019 and economic projection data from 2020 to 2030. The study employed historical and projected population and labor supply from the (UN DESA, 2019) and historical and projected GDP data from the IIASA (2020). The impact of a particular simulation is measured by comparing the baseline and policy simulations. It is important to note that the study only included CO₂ emissions from fossil fuel combustion. Other sources of CO₂ emissions, such as land-use change or deforestation, were taken out.

5 | RESULTS AND DISCUSSION

According to the NDC and National Strategy, Indonesia needs to reduce CO₂ emissions from the energy sector by 18.8% under BAU as part of a bigger plan to reduce the total CO₂ emission by 37.7% in 2030 with international support. Utilizing our dynamic CGE model by simulating the CO₂ emission reduction from the energy sector by 18.8%, we found that the appropriate tax rate scenario amounted to 28.8 USD/ton CO₂ to achieve this CO₂ emission reduction target. This rate is comparable to that of previous work by Yusuf and Resosudarmo (2015), applying a 30 USD/ton CO₂ carbon tax rate and Dissanayake et al. (2020), about 36 USD/ton CO₂.

The study uses the CGE model and estimates four simulations to determine the impact of the carbon tax implementation on macroeconomic indicators and sectoral implications, including the power plant types' usage, environmental indicators, and energy mix. These four simulations are summarized in Table 1. Simulation 1 assumes that carbon tax yield is kept as government saving instead of distributed to the economy. The last three simulations present different types of recycling scenarios. Simulation 2 assumes that the carbon tax yield is fully transferred to households. Simulations 3a and 3b assume that only half of the carbon tax yield went into a lump-sum household transfer. In contrast, the other half is spent on investments, infrastructure, and renewable energy, respectively.

Table 2 shows the impact of levying carbon tax on selected macroeconomic indicators. Simulations 1 and 2 show a negative effect on GDP and real consumption compared to the BAU since carbon tax will increase fossil energy costs and later production costs, impacting producers and consumers. Subsequently, demand for products will decrease in response to rising product prices. This condition also explains the negative impact on employment in both scenarios since

firms will reduce labor input to deal with decreasing product demand. However, Simulation 2 demonstrates a better result than Simulation 1 in terms of overall macroeconomic indicators. Transferring the entire carbon tax yield to households will encourage consumption. Hence, real consumption and GDP will be pushed back up. The long-term adverse effect on GDP and real consumption shown by Simulations 1 and 2 are getting worse, though with different magnitudes. With the tax recycling policy shown in Simulation 2, the negative effect of imposing a carbon tax could be lessened since the tax revenue is allocated to replace the household burden with the lump-sum transfer. In terms of employment, Simulations 1 and 2 show different patterns than GDP or real consumption. Firms will undergo negative impacts from higher energy prices in the short term, indicated by a relatively large fall in employment compared to the baseline. However, employment will start to recover in the medium term as the model assumes full employment in the long run.

Compared to Simulation 2, Simulations 3a and 3b show a relatively better effect on GDP. When half of the tax revenue is reallocated to investment, the multiplier effects of investment will lessen the carbon tax burden, as it boosts production activity starting in the medium term. In terms of real consumption, Simulations 3a and 3b have worse results compared to Simulation 2 at the beginning of the period. It may happen as in Simulations 3a and 3b; the recycling funds are not fully transferred to households, resulting in a more massive decrease in real consumption. Next, Simulation 3a shows positive effects on GDP and employment in the medium-long run since investment in infrastructure will yield a more significant multiplier effect than the other simulations, considerably increasing economic production activity. In terms of real consumption, Simulation 3a shows a decrease in the short run but recovers and gain a higher rate than the baseline in the long run. This condition is also positively impacting employment. Compared to Simulation 3a, Simulation 3b has the worse macroeconomic impact. Although investment in renewable energy could stimulate economic growth, the effect might be less significant and yields smaller benefits to the household than investment in infrastructure. Thus, investing part of carbon tax revenue in infrastructure could have a better impact than all the other scenarios.

Several factors could explain the unprogressive impact on real household consumption in Simulations 2 and 3a. Firstly, the higher price of energy due to carbon tax led to the higher saving demand of energy consumption, as revealed by the research of Burke and Kurniawati (2018) when the Gol implemented electricity subsidy removal. Secondly, it is also supported by a small proportion of household energy expenditure. A study by Hartono, Hastuti, et al. (2020), Hartono, Komarulzaman, et al. (2020) found that nationally, energy expenditure was around 5% in 2018 compared to 7% in 2008. Specifically, on electricity expenditure when the Gol implemented subsidy removal during 2013–2015, the total electricity expenditure was stable from 2008 (before subsidy removal) to 2018 (after subsidy removal), 2.6% and 2.8%, respectively, thereby the carbon tax might drive energy saving and efficiency.

Table 3 shows that imposing carbon tax impacts varies across sectors, even though most sectors will be negatively affected by the

TABLE 2 Impacts of imposing a carbon tax on macroeconomic indicators (cumulative percentage change).

	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
SIM1										
GDP	−0.72	−0.72	−0.76	−0.83	−0.93	−1.05	−1.19	−1.33	−1.47	−1.60
Real consumption	−2.56	−2.93	−3.08	−3.19	−3.29	−3.40	−3.53	−3.68	−3.84	−4.00
Employment	−1.02	−0.84	−0.65	−0.50	−0.40	−0.34	−0.31	−0.30	−0.30	−0.30
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
SIM2										
GDP	−0.64	−0.61	−0.62	−0.66	−0.73	−0.82	−0.91	−1.02	−1.12	−1.21
Real consumption	0.17	−0.18	−0.25	−0.29	−0.32	−0.36	−0.43	−0.52	−0.61	−0.71
Employment	−0.91	−0.73	−0.54	−0.40	−0.30	−0.24	−0.22	−0.21	−0.20	−0.20
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
SIM3a										
GDP	−0.43	−0.23	−0.05	0.16	0.41	0.68	0.96	1.26	1.56	1.79
Real consumption	−1.34	−1.59	−1.79	−1.70	−1.43	−1.13	−0.91	−0.76	−0.58	−0.64
Employment	−1.30	−0.91	−0.66	−0.37	−0.10	0.11	0.23	0.29	0.33	0.20
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
SIM3b										
GDP	−0.58	−0.44	−0.45	−0.52	−0.60	−0.69	−0.78	−0.86	−0.94	−1.00
Real consumption	−0.74	−0.89	−1.12	−1.27	−1.38	−1.48	−1.59	−1.69	−1.77	−1.84
Employment	−0.71	−0.48	−0.47	−0.42	−0.36	−0.31	−0.27	−0.24	−0.21	−0.19

Source: Hartono et al.'s calculation (2020).

policy. As expected, the negative sectoral effects are more significant in Simulation 1 compared to the other three simulations. The most notable finding in Table 3 is that the carbon tax implementation will severely affect the utility sector in all simulations. Electricity is a crucial part of the utility sector and the highest contributor to emissions, so the carbon tax will increase the price of energy. There are at least three more notable findings from these results. Firstly, the total allocation of carbon tax revenue to the household will have a lower impact on the utility sector, but the rate is not significantly different from other simulations. Secondly, a carbon tax only without a recycling policy has the highest negative impact at almost all simulation times. Third, although Simulation 3 has a relatively high contracting effect on the utility sector, the impact on the other sectors is minimal and tends to be positive in the medium and long term. Thus, considering the overall effect, the allocation of carbon tax revenue on infrastructure investment might be a better option to press the negative impact of a carbon tax, following Simulation 3a compared to all other simulation results.

Through expectations, in Table 4, carbon tax implementation could favor environmental indicators, namely CO₂ emission and energy intensity. All simulations generally have lower CO₂ emissions (ranging from 23% to 43%) than the BAU condition, implying significantly less environmental pressure. Also, all simulations result above the government's 18.8% target sets. Simulation 3b shows the best result as expected, and Simulation 2 has a relatively less CO₂ emission reduction. The rebound effect from household consumption in Simulation 2 seems to encourage energy-intensive sectors to increase their production.

In terms of energy intensity, all the simulations imply that carbon tax implementation will significantly reduce energy intensity. Energy intensity is expected to be 16.4% lower than BAU conditions in 2021 when the government reallocates the carbon tax revenue to a mix of household transfer and renewable energy investment (SIM3b). This effect will improve further up to 27.1% in 2030. These results show that Simulation 3b is superior to all other simulations regarding environmental indicators. Since energy intensity is derived by dividing energy consumption by GDP, energy intensity could decrease if the GDP growth rate is faster than the energy consumption. Simulation 3b, based on the additional calculation, shows that GDP grows at 5.98% while energy consumption is at 0.6%.

Table 5 shows the impact of imposing a carbon tax on energy use. Generally, in all simulations, a carbon tax increases renewable energy-based consumption and decreases non-renewable energy consumption, that is, coal and oil. Having 62.4 trillion cubic feet of natural gas reserves, with 43.6 trillion cubic feet being proven, the country's natural gas reserves will last for the next 20 years (MEMR, 2021). This estimation is based on the assumption of no discovery and the production level of 6 billion standard cubic feet per day (BSCFD). The country needs to attract substantive investment for natural gas infrastructure as well as upstream gas exploration and exploitation of new gas fields to fulfill future demand (Purwanto et al., 2016).

Coal energy-based is highly impacted with a considerable reduction, 77%–89%, followed by oil-based energy, 3%–7%. Meanwhile, renewable energy could increase substantially from 38% to 190%. Without renewable energy investment, a carbon tax still reduces coal

TABLE 3 The sectoral impacts of levying carbon tax (cumulative percentage change).

	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
SIM1										
Agriculture	-0.22	-0.13	-0.11	-0.13	-0.18	-0.26	-0.36	-0.47	-0.59	-0.71
Mining	-0.13	0.01	0.08	0.11	0.12	0.10	0.05	-0.02	-0.10	-0.19
Manufacture	-0.97	-0.81	-0.78	-0.81	-0.90	-1.02	-1.16	-1.29	-1.42	-1.54
Utility	-20.88	-22.06	-21.84	-21.66	-21.54	-21.47	-21.42	-21.37	-21.32	-21.27
Construction	-1.16	-1.88	-2.30	-2.57	-2.77	-2.96	-3.14	-3.29	-3.42	-3.51
Trade transport	-1.35	-1.41	-1.54	-1.72	-1.93	-2.16	-2.37	-2.56	-2.71	-2.83
Other services	-0.82	-0.89	-1.04	-1.26	-1.52	-1.80	-2.07	-2.32	-2.54	-2.73
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
SIM2										
Agriculture	-0.31	-0.21	-0.17	-0.16	-0.16	-0.18	-0.20	-0.24	-0.27	-0.31
Mining	-0.51	-0.44	-0.50	-0.60	-0.73	-0.87	-1.02	-1.15	-1.28	-1.40
Manufacture	-1.26	-1.16	-1.19	-1.29	-1.43	-1.59	-1.75	-1.89	-2.01	-2.12
Utility	-18.71	-20.26	-19.95	-19.69	-19.49	-19.35	-19.24	-19.14	-19.05	-18.97
Construction	-0.77	-1.27	-1.55	-1.73	-1.87	-2.01	-2.16	-2.29	-2.40	-2.49
Trade Transport	-0.84	-0.86	-0.91	-1.00	-1.13	-1.28	-1.42	-1.55	-1.66	-1.76
Other services	-0.50	-0.48	-0.48	-0.53	-0.60	-0.70	-0.80	-0.90	-1.00	-1.09
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
SIM3a										
Agriculture	-0.11	0.10	0.30	0.51	0.75	1.01	1.30	1.61	1.92	2.21
Mining	-0.25	-0.09	0.05	0.21	0.39	0.61	0.87	1.16	1.46	1.82
Manufacture	-1.03	-0.84	-0.66	-0.49	-0.30	-0.07	0.19	0.49	0.81	1.20
Utility	-19.81	-21.03	-21.65	-21.78	-21.65	-21.40	-21.14	-20.88	-20.57	-20.40
Construction	0.68	1.20	1.65	2.37	3.21	4.04	4.72	5.23	5.60	4.87
Trade Transport	-0.95	-0.84	-0.73	-0.55	-0.32	-0.06	0.21	0.47	0.74	0.93
Other services	-0.61	-0.56	-0.54	-0.50	-0.42	-0.29	-0.13	0.07	0.30	0.55
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
SIM3b										
Agriculture	-0.27	-0.22	-0.26	-0.33	-0.40	-0.46	-0.51	-0.56	-0.59	-0.63
Mining	-0.43	-0.44	-0.53	-0.66	-0.81	-0.94	-1.06	-1.16	-1.24	-1.32
Manufacture	-1.20	-1.15	-1.20	-1.32	-1.46	-1.60	-1.71	-1.80	-1.86	-1.92
Utility	-19.70	-21.33	-21.29	-21.27	-21.32	-21.38	-21.42	-21.41	-21.36	-21.29
Construction	0.34	0.39	-0.20	-0.54	-0.74	-0.87	-0.99	-1.09	-1.20	-1.29
Trade Transport	-0.96	-0.95	-1.08	-1.24	-1.40	-1.55	-1.69	-1.79	-1.87	-1.93
Other services	-0.63	-0.65	-0.75	-0.90	-1.05	-1.20	-1.34	-1.46	-1.56	-1.64

Source: Author's (2020) calculation.

Note: The study aggregates the 63 sectors into seven sectors for straightforward analysis.

and oil-based energy since it will increase the cost of producing fossil energy. With a carbon tax, the price of new and renewable energy can compete with fossil energy and carbon-based fuels. Industry, business, household, and electricity reduce their consumption of non-renewable energy by consuming more energy-efficient products and technologies or shifting their consumption to relatively cheaper energy. The significant decrease in coal consumption followed by a substantial increase in renewable energy usage is in line with the current Gol plan that intends to decommission the coal power plant in addition to

doubling the current capacity of the renewable energy power plant by 2028. Research examining the socioeconomic impacts of levying a carbon tax in China using a similar approach found that a higher tax on coal-related fuels is more efficient than a conventional carbon tax policy (Fu et al., 2021).

Among the simulations, Simulation 3b shows an enormous positive impact on renewable energy use, that is, up to 190%, and the largest negative impact on non-renewable energy. Investing part of carbon tax revenue in renewable energy could decrease the price due

TABLE 4 The impacts of imposing a carbon tax on environmental indicators (cumulative percentage change).

CO ₂ Emission	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
SIM1	-26.5	-28.4	-27.9	-27.5	-27.2	-27.0	-26.8	-26.7	-26.6	-26.6
SIM2	-24.4	-26.6	-25.8	-25.1	-24.5	-24.1	-23.7	-23.4	-23.3	-23.2
SIM3a	-25.5	-27.4	-28.9	-30.0	-30.7	-31.3	-31.8	-32.3	-32.8	-33.5
SIM3b	-25.5	-31.7	-36.0	-38.3	-39.7	-40.6	-41.4	-42.0	-42.5	-43.0
Energy intensity	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
SIM1	-17.5	-19.7	-18.2	-16.7	-15.2	-14.0	-13.0	-12.1	-11.3	-10.7
SIM2	-14.9	-17.6	-15.5	-13.5	-11.7	-10.2	-8.8	-7.7	-6.8	-6.2
SIM3a	-16.5	-18.9	-20.8	-22.0	-22.8	-23.4	-24.0	-24.5	-25.0	-25.6
SIM3b	-16.4	-22.4	-25.3	-26.2	-26.7	-26.8	-27.0	-27.0	-27.1	-27.1

Source: Author's (2020) calculation.

TABLE 5 Long-run impacts of imposing a carbon tax on energy use.

	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
<i>SIM-1</i>										
Coal	-78.92	-80.08	-80.83	-81.58	-82.28	-82.89	-83.40	-83.80	-84.12	-84.37
Oil	-4.89	-5.21	-5.54	-5.88	-6.21	-6.52	-6.79	-7.01	-7.17	-7.27
Gas	38.52	31.03	40.90	50.50	59.30	67.06	73.74	79.45	84.25	88.16
Renewable	38.67	40.96	42.83	44.91	46.89	48.64	50.16	51.50	52.69	53.74
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
<i>SIM-2</i>										
Coal	-77.43	-78.89	-79.71	-80.52	-81.26	-81.90	-82.43	-82.86	-83.20	-83.48
OIL	-3.79	-4.11	-4.41	-4.72	-5.02	-5.30	-5.53	-5.72	-5.85	-5.94
Gas	46.50	37.60	50.10	62.13	73.13	82.82	91.11	98.08	103.80	108.34
Renewable	40.33	42.45	44.10	45.98	47.80	49.44	50.90	52.20	53.37	54.41
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
<i>SIM-3a</i>										
Coal	-78.19	-79.43	-80.38	-81.00	-81.42	-81.72	-81.98	-82.19	-82.35	-82.57
OIL	-4.34	-4.51	-4.70	-4.64	-4.44	-4.22	-4.06	-3.98	-3.93	-4.22
Gas	42.37	34.64	28.18	23.68	20.52	18.13	16.08	14.30	12.92	11.14
Renewable	39.50	41.86	45.80	50.57	55.63	60.64	65.37	69.75	73.78	77.32
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
<i>SIM-3b</i>										
Coal	-78.44	-83.02	-85.99	-87.45	-88.28	-88.81	-89.17	-89.46	-89.68	-89.87
OIL	-3.81	-4.03	-4.45	-4.71	-4.85	-4.94	-5.00	-5.05	-5.09	-5.12
Gas	41.03	8.66	-13.29	-23.91	-29.92	-33.67	-36.29	-38.33	-40.00	-41.33
Renewable	40.73	77.54	118.10	142.76	158.37	168.71	176.13	181.93	186.68	190.51

Source: Author's (2020) calculation.

to economies of scale, making renewable energy relatively cheaper than non-renewable energy. Thus, people are eager to shift their energy consumption from non-renewable to renewable ones. Hence, in Simulation 3b, renewable (non-renewable) energy consumption dramatically increases (decreases).

In this study, the sensitivity analysis will refer to Yusuf and Resosudarmo (2015) and Warr and Yusuf (2011). The present study applies changes in 4 parameter values, including (i) σ_{CE} ($SIGMA_{CE}$), the CES

capital-energy elasticities; (ii) σ_{E1} ($SIGMA_{E1}$), the CES substitution among energy types; (iii) σ_{E2} ($SIGMA_{E2}$); the CES substitution among fuel energy types; and (iv) σ_{E3} ($SIGMA_{E3}$); the CES substitution among diesel energy types to examine the changes in the endogenous variables. The standard parameter from each elasticity value used in this study are σ_{CE} ($SIGMA_{CE}$) = 0.5, σ_{E1} ($SIGMA_{E1}$) = 0.3, σ_{E2} ($SIGMA_{E2}$) = 3, and σ_{E3} ($SIGMA_{E3}$) = 0.3. Specifically for the sensitivity analysis, changes are applied to 4 parameters, following the value changes in Table 6.

Type of parameter	Bottom value	Standard value	Top value
CES capital-energy elasticities (σ_{CE})	0.375	0.50	0.625
CES substitution among energy types (σ_{E1})	0.225	0.30	0.375
CES substitution among fuel energy types (σ_{E2})	2.250	3.00	3.750
CES substitution among diesel energy types (σ_{E3})	0.225	0.30	0.375

Note: The bottom value represents 75% of the standard value, the top value represents 125% of the standard value, and the lower limit of the parameter σ_{CE} for Sim2 and Sim3b is 0.4.

TABLE 6 Changes parameters for sensitivity analysis.

TABLE 7 Value changes in macroeconomic variables.

Variable		Parameter								
		Standard	σ_{CE}		σ_{E1}		σ_{E2}		σ_{E3}	
			Bottom	Top	Bottom	Top	Bottom	Top	Bottom	Top
GDP	SIM1	−1.60	−1.60	−1.63	−1.62	−1.59	−1.60	−1.60	−1.61	−1.60
	SIM2	−1.21	−1.21	−1.24	−1.23	−1.20	−1.21	−1.21	−1.22	−1.20
	SIM3a	1.79	1.83	1.73	1.80	1.80	1.79	1.79	1.79	1.79
	SIM3b	−1.00	−0.99	−0.98	−1.04	−0.97	−1.00	−1.00	−1.01	−0.99
Real consumption	SIM1	−4.00	−4.06	−4.07	−4.09	−3.98	−4.00	−4.00	−4.02	−3.98
	SIM2	−0.71	−0.80	−0.77	−0.80	−0.69	−0.71	−0.71	−0.74	−0.69
	SIM3a	−0.64	−0.72	−0.62	−0.69	−0.58	−0.64	−0.64	−0.64	−0.63
	SIM3b	−1.84	−1.77	−1.90	−1.94	−1.77	−1.84	−1.85	−1.87	−1.82
Employment	SIM1	−0.30	−0.30	−0.30	−0.31	−0.30	−0.30	−0.30	−0.30	−0.30
	SIM2	−0.20	−0.21	−0.21	−0.21	−0.20	−0.20	−0.20	−0.21	−0.20
	SIM3a	0.20	0.21	0.19	0.20	0.20	0.20	0.20	0.20	0.20
	SIM3b	−0.19	−0.17	−0.20	−0.20	−0.18	−0.18	−0.19	−0.19	−0.18
Sectoral—agriculture	SIM1	−0.71	−0.59	−0.79	−0.74	−0.67	−0.70	−0.71	−0.71	−0.70
	SIM2	−0.31	−0.22	−0.40	−0.35	−0.27	−0.31	−0.31	−0.32	−0.30
	SIM3a	2.21	2.34	2.07	2.17	2.25	2.21	2.21	2.21	2.21
	SIM3b	−0.63	−0.56	−0.70	−0.66	−0.61	−0.62	−0.63	−0.63	−0.62
Sectoral—mining	SIM1	−0.19	0.08	−0.40	−0.16	−0.21	−0.19	−0.19	−0.20	−0.18
	SIM2	−1.40	−1.19	−1.60	−1.37	−1.42	−1.40	−1.40	−1.40	−1.39
	SIM3a	1.82	2.04	1.59	1.86	1.79	1.82	1.82	1.82	1.82
	SIM3b	−1.32	−1.19	−1.49	−1.30	−1.35	−1.32	−1.32	−1.33	−1.32
Sectoral—manufacture	SIM1	−1.54	−1.24	−1.79	−1.58	−1.49	−1.54	−1.54	−1.55	−1.52
	SIM2	−2.12	−1.89	−2.36	−2.16	−2.08	−2.12	−2.12	−2.14	−2.10
	SIM3a	1.20	1.44	0.95	1.16	1.24	1.20	1.20	1.18	1.21
	SIM3b	−1.92	−1.73	−2.13	−1.98	−1.86	−1.92	−1.92	−1.93	−1.90
Sectoral—utility	SIM1	−21.27	−20.27	−23.27	−22.00	−21.45	−21.27	−21.28	−21.01	−21.56
	SIM2	−18.97	−18.63	−20.73	−19.66	−19.12	−18.97	−18.98	−18.71	−19.26
	SIM3a	−20.40	−17.74	−22.23	−20.81	−20.18	−20.40	−20.40	−19.94	−20.85
	SIM3b	−21.29	−18.45	−24.51	−22.78	−18.79	−21.28	−21.29	−21.17	−21.41
Sectoral—construction	SIM1	−3.51	−3.76	−3.41	−3.59	−3.50	−3.51	−3.51	−3.53	−3.49
	SIM2	−2.49	−2.71	−2.39	−2.56	−2.48	−2.49	−2.49	−2.52	−2.47
	SIM3a	4.87	5.02	4.73	4.85	4.90	4.88	4.87	4.87	4.88
	SIM3b	−1.29	−1.25	−1.24	−1.36	−1.25	−1.29	−1.29	−1.31	−1.27
Sectoral-trade transport	SIM1	−2.83	−2.73	−2.98	−2.90	−2.80	−2.83	−2.84	−2.85	−2.82
	SIM2	−1.76	−1.69	−1.90	−1.82	−1.73	−1.76	−1.76	−1.78	−1.74
	SIM3a	0.93	1.04	0.80	0.89	0.98	0.93	0.93	0.93	0.94
	SIM3b	−1.93	−1.81	−2.06	−2.01	−1.88	−1.93	−1.94	−1.95	−1.91

TABLE 7 (Continued)

Variable		Parameter								
		Standard	σ_{CE}		σ_{E1}		σ_{E2}		σ_{E3}	
			Bottom	Top	Bottom	Top	Bottom	Top	Bottom	Top
Sectoral—other services	SIM1	−2.73	−2.68	−2.84	−2.78	−2.72	−2.73	−2.74	−2.75	−2.72
	SIM2	−1.09	−1.06	−1.19	−1.14	−1.07	−1.09	−1.09	−1.11	−1.07
	SIM3a	0.55	0.60	0.47	0.52	0.58	0.55	0.55	0.54	0.55
	SIM3b	−1.64	−1.55	−1.72	−1.70	−1.60	−1.64	−1.64	−1.66	−1.63
CO ₂ Emission	SIM1	−26.57	−27.85	−28.89	−25.71	−29.01	−26.67	−26.46	−26.15	−27.00
	SIM2	−23.18	−24.62	−25.15	−22.43	−25.36	−23.28	−23.08	−22.80	−23.58
	SIM3a	−33.46	−32.32	−34.47	−31.03	−35.82	−33.49	−33.44	−33.30	−33.63
	SIM3b	−42.99	−42.00	−44.30	−39.87	−45.33	−43.13	−42.84	−42.93	−43.05
Energy intensity	SIM1	−10.73	−13.57	−13.96	−11.33	−13.14	−10.81	−10.64	−10.02	−11.45
	SIM2	−6.16	−9.11	−8.81	−6.98	−8.11	−6.24	−6.08	−5.52	−6.84
	SIM3a	−25.61	−24.24	−26.78	−23.98	−27.68	−25.63	−25.59	−25.33	−25.89
	SIM3b	−27.14	−25.13	−29.66	−25.08	−28.77	−27.26	−27.02	−27.05	−27.23
Energy mix—coal	SIM1	5.78	6.55	5.57	7.95	4.16	5.79	5.78	5.80	5.77
	SIM2	5.81	6.43	5.58	7.93	4.22	5.81	5.80	5.83	5.79
	SIM3a	7.39	7.64	7.19	9.90	5.40	7.39	7.38	7.48	7.29
	SIM3b	4.60	4.72	4.37	7.17	2.87	4.60	4.59	4.64	4.55
Energy mix—oil	SIM1	42.29	44.12	43.24	42.64	43.34	42.23	42.36	41.93	42.67
	SIM2	40.74	42.36	41.36	41.15	41.50	40.68	40.80	40.43	41.07
	SIM3a	50.00	49.66	50.28	49.16	51.21	49.99	50.02	49.75	50.26
	SIM3b	53.03	52.09	54.30	51.66	54.20	52.96	53.10	52.95	53.12
Energy mix—gas	SIM1	40.27	36.73	39.35	38.18	40.10	40.32	40.22	40.72	39.80
	SIM2	42.34	39.33	41.85	40.19	42.55	42.39	42.29	42.71	41.94
	SIM3a	27.23	27.17	27.28	26.68	27.01	27.23	27.22	27.35	27.10
	SIM3b	15.39	15.75	14.92	16.66	13.84	15.40	15.37	15.45	15.31
Energy mix—renewable	SIM1	11.65	12.59	11.84	11.22	12.40	11.67	11.64	11.55	11.76
	SIM2	11.11	11.88	11.21	10.74	11.74	11.12	11.10	11.04	11.20
	SIM3a	15.38	15.53	15.25	14.27	16.38	15.39	15.38	15.42	15.35
	SIM3b	26.99	27.45	26.41	24.51	29.09	27.04	26.93	26.97	27.01

The results from the four parameter value changes on macroeconomic variables (GDP, real consumption, and employment), sectoral analysis, and energy indicators (CO₂ emission, energy intensity, and energy mix) are reported in Table 7. In general, the sensitivity analysis results show that the changes in endogenous variables (macroeconomic variables, sectoral analysis, and energy indicators) are not sensitive to parameter value changes (variation in parameter values) related to energy elasticity. Therefore, the results of the simulations conducted in this paper are relatively robust.

6 | CONCLUSION

This article examines the simulation of the role of carbon taxes in Indonesia. The carbon tax is a tool to achieve the RUEN and INDC

targets. As Indonesia's commitment to participate in ratifying the Paris Agreement, INDC has been in line with the targets for achieving pillars 7 and 13 of the SDGs. Specifically, this study analyzes the impact of carbon taxes on environmental and macroeconomic indicators. The model utilizes a recursive dynamic CGE model with detailed energy sectors and sets up four different scenarios in the simulation. In comparison, many other studies rely on energy aggregate data.

The study highlights some important results that the Gol should address. Firstly, the simulation shows that to achieve the CO₂ emission reduction target from the energy sector by 18.8% under BAU, the Gol needs to impose a tax rate of up to 28.8 USD/ton of CO₂. But, levying the carbon tax contracts the economy. It reduces GDP, real consumption, employment rate, and sectoral output, particularly for the utility sector, which heavily depends on electricity. Secondly, carbon tax revenue recycling could reduce the contracting impact of

the carbon tax application. The results have shown that carbon tax revenue recycling on a combination of household transfer and investments (Simulations 3a and 3b) yields better results than carbon tax revenue recycling only on household transfer (Simulation 2) or alone without transfer to the household (Simulation 1). Carbon tax revenue recycling on a mix of household transfer and infrastructure investment generates a higher GDP, higher employment rate, and less negative impact on household consumption. Thirdly, a carbon tax revenue recycling on household transfer and RE investment is a better recycling option for greener energies with more considerable CO₂ emission reduction, 20.9% or slightly above the target in 2030 (18.8%). Four, the carbon tax policy could boost a higher energy mix by inducing RE consumption and reducing coal and oil (non-RE) consumption in all simulations. Among the simulations, Simulation 3b provides the best RE used with the possibility of up to 190% increase in RE. Albeit Simulation 3a and Simulation 3b reflect different priorities that seemed like trade-offs for each other, investment diversifications between infrastructure and RE could aim at emission reduction and economic enhancement. If the non-RE consumption proportion does not change but the RE investment increase, followed by the increase in RE consumption, then the RE mix will potentially increase.

Given that many countries growingly adopt a carbon tax, the study recommends several policy options to implement the tax in a better way, that is, (i) the Gol should prioritize a carbon tax on fossil fuel-based energy, mainly on electricity generators and large industrial firms since both sources have more significant emissions compared to other sources as it is more efficient (Ministry of Finance, 2009); (ii) it is crucial to set a tax price at a level that will not cause shocks in production costs for firms and price shocks for households as carbon tax might draw setbacks on macroeconomic indicators, such as GDP, employment, household consumption, and output sectoral; (iii) the Gol should apply a transitional policy to promote energy reform, such as levying the tax gradually and reinvest the revenue on RE immediately. In this way, fossil fuel-based energy dependency could be reduced, and the RE supply could be increased; (iv) to reduce emissions while still maintaining macroeconomic indicators, the Gol should diversify the investment strategy on infrastructure and renewable energy.

Our simulations show that imposing a carbon tax could simultaneously contribute to achieving the Paris Agreement and several SDG indicators. The reduction of CO₂ emission is in line with SDG indicator 13.2.2. Meanwhile, the considerable shift from coal and oil to renewable energy could contribute to the achievement of SDGs indicator 7.2.1. Lastly, the reduction in energy intensity is in accordance with SDG indicator 7.3.1.

However, this study has potential limitations. Even though the result could explain the comparable carbon tax rate, it cannot show the impact of a carbon tax on RE mix achievement, which is also targeted in the RUEN. Then, the research does not provide the detail of RE that significantly impacts the reduction of emissions with little adverse impact on the economy. Thus, future research simulating the effect of a carbon tax on the RE mix target and the detailed RE is essential for more information and recommendation on achieving the RE mix target and increasing the RE supply.

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ENDNOTES

¹ see Horridge (2002) for mathematical expressions and illustrations.

² Capital-energy nesting and energy aggregate nesting from standard ORANIG-RD is modified to perform the analysis.

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